n8n

What it does in the Ads Management System

11 functions explained in plain language

n8n is the automation engine that connects all platforms in this system.
Without n8n, every platform — Shopify, Zendesk, Zoho Books, Meta — sits in isolation.
n8n watches for things to happen across all platforms, then takes action automatically.
It runs 24 hours a day, 7 days a week on a Hetzner cloud server.
It handles all brands simultaneously from a single installation.

What is n8n?

n8n (pronounced 'n-eight-n') is a workflow automation platform. It works by listening for events — things that happen in your platforms — and responding to them automatically according to rules you define.

Think of it like a very fast, very reliable staff member who:

- Watches all your platforms simultaneously, 24/7
- Never misses a notification
- Follows the exact same process every single time
- Can handle multiple brands at once without getting confused
- Never forgets to send the conversion data to Meta

In this system, n8n has 11 specific functions. Each one is triggered by something happening in one of your platforms, and each one takes a specific action as a result.

SIMPLE PICTURE OF HOW n8n SITS IN THE SYSTEM:

Shopify redirect page fires → n8n receives it → stores session token
Zendesk ticket created → n8n finds it → writes tracking data into it
Zendesk detects RFQ → n8n creates Zoho Books estimate → emails sales team
Zoho Books Sales Order created → n8n receives it → sends revenue data to Meta CAPI
Every morning → n8n runs health check → emails audit report

Function 1

F1 Receives the click log from Shopify redirect pages

Triggered by	Customer clicks the WhatsApp, Viber, or Messenger contact button on any brand Shopify store
What it does	• The redirect page on Shopify fires a silent message to n8n before opening WhatsApp/Viber/Messenger • n8n receives a package of data containing: session token, fbclid (Meta ad click ID), which channel was clicked, which brand store it came from, and the exact timestamp • n8n stores this data temporarily, waiting for the matching Zendesk ticket to appear • This happens in under 200 milliseconds — the customer notices nothing
Why it matters	*This is the starting point of all attribution. Without capturing this data the moment the customer clicks, there is nothing to link back to the original Meta ad click.*

Function 2

F2 Matches the click log to the Zendesk ticket and enriches the ticket

Triggered by	A new Zendesk ticket is created when the customer sends their first message on WhatsApp, Viber, Messenger, or Email
What it does	• n8n finds the Zendesk ticket that belongs to the customer who just clicked the contact button • The matching method depends on the channel: • WhatsApp → matches using the customer phone number from the Zendesk user profile • Messenger → matches using the customer Facebook Page Scoped ID (PSID) from Zendesk identities • Viber → matches using Viber user ID plus a 30-minute time window • Email → extracts session token directly from the email body using a pattern search • Once matched, n8n writes the session token, fbclid, UTM parameters, channel, and brand into the ticket's custom fields • The ticket is now fully linked back to the original Meta ad click
Why it matters	*Without this step, the Zendesk ticket is just a message with no connection to the ad that brought the customer. This step makes every inquiry traceable.*

Function 3

F3 Retries unmatched tickets every 5 minutes

Triggered by	Scheduled automatic check — runs every 5 minutes around the clock
What it does	• n8n searches for Zendesk tickets created in the last 10 minutes that are still missing a session token • For each unmatched ticket, n8n checks stored customer identifiers — phone numbers, PSIDs, Viber IDs — to see if a match can now be made • For Viber and Messenger, n8n also checks stored pending click logs from within the same 30-minute window • For email tickets, n8n re-attempts the regex extraction from the ticket body • Any successful match enriches the ticket and removes it from the unmatched queue
Why it matters	*Sometimes the Zendesk ticket appears before n8n finishes processing the click log, or there is a slight delay. This function ensures no inquiry is permanently missed due to timing.*

Function 4

F4 Detects RFQ intent and automatically creates a Zoho Books estimate

Triggered by	The rfq_detected tag is added to a Zendesk ticket (triggered by customer using keywords like 'quote', 'magkano', 'how much', 'estimate')
What it does	• n8n detects the rfq_detected tag on the Zendesk ticket • n8n reads all tracking data from the ticket: session token, fbclid, brand, channel, customer name • n8n checks if an estimate already exists for this session token — if yes, skips creation (see Function 5) • n8n creates a blank estimate shell in the correct brand's Zoho Books organization • The estimate is already pre-filled with: session token, fbclid, UTM campaign, contact channel, Zendesk ticket ID, brand, and customer contact details • n8n sends an email to the sales team: New RFQ received, with the Zendesk ticket link and Zoho Books estimate link • The sales agent simply opens the estimate, adds products and prices, and sends it to the customer
Why it matters	*Without this, sales agents would have to manually create every estimate and manually enter all tracking data. One missed field breaks the entire attribution chain. This function makes the process automatic and error-free.*

Function 5

F5 Prevents duplicate Zoho Books estimates

Triggered by	Runs as part of Function 4 — before creating any estimate
What it does	• Before creating a Zoho Books estimate, n8n first searches Zoho Books for any existing estimate that has the same session token in its custom fields • If an estimate already exists for that session token, n8n skips creation entirely • n8n adds an internal note to the Zendesk ticket linking to the existing estimate instead • If no existing estimate is found, n8n proceeds to create a new one
Why it matters	*If a customer says 'can I get a quote' twice in the same conversation, Zendesk would tag the ticket twice with rfq_detected. Without this check, two estimates would be created for the same inquiry — confusing for the sales team and duplicating the tracking data.*

Function 6

F6 Receives the Sales Order webhook from Zoho Books

Triggered by	A Sales Order is created in Zoho Books (staff converts a paid quotation to a Sales Order)
What it does	• Zoho Books automatically sends a notification to n8n the moment a Sales Order is created • n8n receives the full Sales Order data including all custom tracking fields: session token, fbclid, UTM campaign, channel, brand • n8n also receives the financial data: total amount, currency (PHP) • n8n validates the data — if session token is empty or amount is zero, it stops and logs the gap • If validation passes, n8n proceeds to Function 7
Why it matters	*This is the moment the offline sale becomes visible to the digital system. The Zoho Books Sales Order is the final confirmation that a real paid transaction occurred. n8n receiving this is what triggers the Meta CAPI event.*

Function 7

F7 Sends the conversion event to Meta CAPI

Triggered by	Immediately after Function 6 validates the Sales Order data
What it does	• n8n retrieves the customer's email address and phone number from Zoho Books • n8n normalizes the data: email is converted to lowercase with spaces removed, phone is formatted as +63XXXXXXXXX • n8n hashes the email and phone using SHA-256 encryption — this converts them into scrambled codes so Meta never sees the actual personal information • n8n constructs the fbc parameter: fb.1.[timestamp].[fbclid] — this is the field that links the conversion back to the specific ad click • n8n sends a Purchase event to Meta CAPI with: event name (Purchase), sale amount in PHP, hashed email, hashed phone, fbclid, unique event ID • Meta receives this event and learns: this specific ad click led to a sale worth ₱ [amount] • n8n logs the result (success or failure) to the daily audit
Why it matters	*This is the entire point of the system. Meta now knows which ad, which audience, and which creative generated real paid revenue. Over time Meta optimizes your budget toward ads that attract actual buyers — not just people who message and never buy.*

Function 8

F8 Prevents duplicate CAPI events from reaching Meta

Triggered by	Runs as part of Function 7 — on every CAPI event send
What it does	• Every CAPI event n8n sends to Meta includes a unique event_id field: SO-[salesorder_id] • If the same Sales Order webhook is received twice (e.g. due to a Zoho Books retry or n8n restart), n8n would attempt to send the same CAPI event twice • Meta automatically deduplicates events with identical event_id values received within 48 hours — the second event is silently dropped • Additionally, n8n checks its own data store before sending — if the salesorder_id was already processed, it skips the CAPI send entirely
Why it matters	*Sending the same sale to Meta twice would corrupt your revenue data and make your ads appear to generate double the actual revenue. This function ensures every sale is counted exactly once.*

Function 9

F9 Routes all data to the correct brand's systems

Triggered by	On every webhook received — from Shopify, Zendesk, or Zoho Books
What it does	• Every webhook payload that arrives at n8n contains a brand identifier field • n8n reads this brand field and uses it to route the workflow to that brand's specific systems: • → Correct Zoho Books organization (each brand has its own) • → Correct Meta CAPI Dataset ID (each brand has its own) • → Correct email notification recipients (each brand has its own sales team) • A Switch node in n8n handles the routing logic — one output per brand • If an unknown brand is received, n8n logs it as an error and stops — it does not guess
Why it matters	*With 10+ brands running simultaneously on one n8n server, routing is critical. A routing error would send Brand A's customer data to Brand B's Meta ad account — permanently corrupting attribution data for both brands.*

Function 10

F10 Stores customer identifiers for instant repeat-contact matching

Triggered by	After every successful first-contact match in Function 2
What it does	• When n8n successfully matches a customer for the first time, it stores the customer's identifier in its internal data store • What gets stored depends on the channel: • WhatsApp → stores: brand + phone number → session token mapping • Messenger → stores: brand + PSID → session token mapping • Viber → stores: brand + Viber user ID → session token mapping • Email → stores: brand + email address → session token mapping • The brand is included in the storage key so the same customer contacting two different brands gets separate entries per brand • These mappings are stored for 365 days • The next time the same customer contacts the same brand, n8n finds their identifier instantly — no time-window matching needed
Why it matters	*Customers often contact you more than once. A returning customer who clicks a new Meta ad and messages again should have their new ad click attributed correctly. This function makes repeat-customer matching fast and reliable.*

Function 11

F11 Sends the daily audit email every morning

Triggered by	Scheduled automatic trigger — runs every morning at 7:00am Philippines time
What it does	• n8n counts all Zendesk tickets created yesterday that are still missing a session token — these are inquiries where attribution was not captured • n8n counts all Zendesk tickets that were tagged rfq_detected but have no corresponding Zoho Books estimate — these are RFQs the system missed • n8n counts how many Sales Orders were processed and how many CAPI events were successfully sent to Meta • n8n checks the workflow execution log for any failures in the last 24 hours — with error details • n8n sends all of this as a summary email to the monitoring person • The email shows the tracking gap rate: what percentage of yesterday's inquiries have attribution data
Why it matters	*Without daily monitoring, a broken webhook or failed integration could go unnoticed for weeks — silently missing attribution data for dozens of sales. This function surfaces problems every morning so they are fixed quickly.*

All 11 functions — summary

#	Function name	Triggered by	Output
F1	Receives click log from Shopify	Customer clicks WhatsApp/Viber/Messenger button	Session token + fbclid stored in n8n
F2	Matches click log to Zendesk ticket	New Zendesk ticket created	Tracking fields written into ticket
F3	Retries unmatched tickets	Every 5 minutes on schedule	Delayed matches enriched
F4	Detects RFQ and creates estimate	rfq_detected tag on Zendesk ticket	Zoho Books estimate created + sales team emailed
F5	Prevents duplicate estimates	Before every estimate creation	Duplicate estimate skipped if session token already exists
F6	Receives Sales Order webhook	Sales Order created in Zoho Books	Sale data validated and passed to F7
F7	Sends CAPI event to Meta	After F6 validates	Purchase event sent to Meta — revenue attributed to ad
F8	Prevents duplicate CAPI events	On every CAPI send	Duplicate events blocked by event_id deduplication
F9	Routes data by brand	On every webhook received	Data sent to correct brand Zoho Books org and Meta Dataset
F10	Stores customer identifiers	After every successful first match	Phone/PSID/Viber ID stored for instant repeat matching
F11	Sends daily audit email	Every morning 7am	Tracking gaps, failures, and CAPI count reported

The most important function is F7 — sending the CAPI event to Meta.
Everything else exists to support F7 and make sure the data reaching Meta is accurate.
F1 captures the ad click. F2 links it to the inquiry. F4 links it to the quotation.
F6 receives the payment confirmation. F7 tells Meta which ad made the money.
F8, F9, F10 protect data quality. F3, F5, F11 catch mistakes and gaps.